

A Thematic Layer for CLAUDETTE: Separating the Cost of Multi-Label Annotation from the Effect of Taxonomy Granularity

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Abstract. The CLAUDETTE/UNFAIR-ToS corpus is the reference benchmark for unfair clauses in online terms of service, but it records only whether a sentence is unfair, never what the clause is about. We add a thematic layer to its 50 contracts: 9,414 sentences, each labelled with one or more of 20 legal themes independently by three trained annotators, and by four large language models on the same sentences. The layer separates two effects that the reliability literature tends to conflate. Letting annotators assign several themes lowers agreement, by much the same amount whatever the number of categories. Merging the taxonomy from 20 themes into 11 recovers an equivalent amount, and the gain holds on contracts held out from the design of the merges. Merging is constrained, however: the taxonomy that agreement alone would select groups clauses that consumer protection law treats differently, at more than twice the cost in downstream signal. A fine-tuned legal encoder outperforms every language-model judge, while remaining clearly below the human ceiling. We release the annotations with their individual votes, the code-book, the protocol and the annotation platform.

Keywords. unfair terms of service, CLAUDETTE, multi-label annotation, inter-annotator agreement, LLM-as-a-judge, legal language models, annotation platform

1. Introduction

CLAUDETTE records whether a sentence in a consumer contract is potentially unfair; it records nothing about what the clause is about [1,2]. A limitation clause and an arbitration clause are both unfair, and both are simply unfair. Where clause topics do exist, they come without chance-corrected agreement [3,4], so the reliability of the task itself is unknown.

We add a thematic layer to the reference benchmark. All 50 contracts and 9,414 sentences of CLAUDETTE are labelled with one or more of 20 legal themes, independently

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by three trained annotators, and separately by four large language models prompted with the same vocabulary on the same sentences. Individual votes, the arbitration trail and the resulting gold standard are all released.

The layer settles a question that the reliability literature leaves open. Agreement is known to fall when annotators may assign several labels, and to rise when categories are merged [5,6]; what is not known is whether these are one effect or two, since they are seldom measured on the same sentences. Here they are. The multi-label format costs 0.073 of chance-corrected agreement and varies by only 0.012 across four granularities, while merging 20 themes into 11 recovers 0.067 — a gain that replicates on 17 contracts held out from the design of the merges. Consolidation is therefore a real lever on reliability, but not an unconstrained one: the taxonomy that agreement alone would select merges clauses across unfairness strata, and loses 13.2% of the downstream unfairness signal to buy 0.012 of extra agreement.

We contribute: (i) a thematic layer over CLAUDETTE, multi-label and triply annotated, published with its disagreements and with the open-source platform used to produce it; (ii) a separation of the cost of the multi-label *format* from the effect of taxonomy *granularity*, measured on the same sentences and replicated out of sample; and (iii) a legal constraint on consolidation — merges must not cross unfairness strata — together with the price of violating it.

2. Background and Related Work

Automatic detection of potentially unfair clauses in terms of service is by now well established: CLAUDETTE and its successors detect and explain unfairness in English and across languages [1,2,7,8,9,10], and the task is part of standard legal benchmarks [11]. Compact domain-specific encoders remain competitive with, and often better than, general-purpose large models on clause classification [12,13,14,15]. Language models are increasingly used as semantic annotators of legal text, with documented gains in cost and documented risks of anchoring and circularity [16,17,18,19,20], and, as judges, with known divergence from human evaluators [21,22,23]. On the measurement side, chance-corrected agreement for set-valued annotation is a mature subject [24,25,26,6,27], and a meta-analysis of annotation studies identifies the number of categories as one of the strongest determinants of agreement [5].

What is missing is a reliable, measured and published thematic layer on the reference benchmark itself. Contract-topic resources adjacent to ours are either single-annotated [3], defined for a different document type [28,29,30], or reported without chance-corrected agreement [4,31]; none places human annotators and language models on the same vocabulary and the same sentences. The consequence is methodological as much as practical: because the multi-label format and the granularity of the taxonomy are almost never varied on the same data, it is not known whether the agreement penalty attributed to multi-label annotation is a property of the format or of the label set it is applied to — a question [5] raises and leaves open.

3. The Thematic Layer and Its Construction

The layer. We annotate all 50 English contracts of CLAUDETTE [1]: 9,414 sentences, each labelled with one primary theme and zero or more secondary themes drawn from a 20-theme legal vocabulary (T20) covering the recurring subject matter of online terms of service. Every sentence was labelled independently by each of three trained annotators working from a written codebook, and, separately, by four large language models prompted with the same vocabulary on the same sentences. Individual votes are retained rather than collapsed into a consensus [32,33,34].

Construction. Annotators did not post-edit the output of one model. The four judges labelled every sentence independently, and a deterministic engine sorted each sentence into one of five tiers on inter-judge agreement alone, from unanimous to fully split. Annotators saw that tier and the competing candidates, but selected from the full 20-theme vocabulary, and frequently did: 34.5% of gold labels match none of the four proposals, rising to 58.7% on the sentences annotators themselves disputed. Annotators could not see one another’s work. The gold standard is built by a three-tier cascade: 46.8% of sentences are unanimous, 48.3% carry a majority of at least two of three, and the remaining 4.9% (462 sentences) were arbitrated by a human committee. That last share is not overhead but a measurement: it is the fraction of the corpus on which three trained annotators, given the same codebook, cannot converge. Crucially, *model outputs are never parties to the gold*: resolution is strictly inter-annotator, and the judges are held out as an object of evaluation rather than a contributor to the reference [23].

T14 and T11 were designed under four criteria — confusability, reliability, contractual function and support — and one constraint fixed in advance: no merge joins themes whose observed unfairness rates fall in opposite strata. The merges were reviewed by the legal scholars among the authors; T10 breaks that constraint by construction and is reported as a counterfactual.

Measurement. We report Krippendorff’s α under two distances [24,26]: the MASI distance for set-valued labels [25], and the nominal distance on primary themes alone. The gap between them is what multi-label annotation costs in reliability [6]. Coarser taxonomies (T14, T11, T10) are *projections* of T20 applied at read time: the annotations themselves are never rewritten, so every comparison is paired by construction, and confidence intervals come from bootstrap resampling over documents rather than sentences. The merges were designed on 33 contracts and validated on the 17 held out from that design. Finally, the human ceiling is estimated leave-one-annotator-out, each annotator being scored against a gold to which they did not contribute [35].

4. Results

The cost of the multi-label format. Scoring secondary themes lowers agreement from a nominal α of 0.732 to an α -MASI of 0.658: the multi-label format costs 0.073 [0.061, 0.087] of chance-corrected agreement. Across the four granularities it stays between 0.063 and 0.075, without decreasing monotonically as categories are merged (Table 1); its whole range, 0.012, is five times smaller than the agreement gain consolidation yields below. The two effects that the literature tends to conflate are therefore of different orders, and can be measured separately on the same sentences.

Table 1. Reliability and downstream signal by taxonomy, on all 50 contracts (9,414 sentences, three annotators). The gap between α -MASI and nominal α is the cost of the multi-label format. Differences against T20 are paired by document; 95% confidence intervals from 1,000 document-level bootstrap resamples. Signal loss is the relative drop in leave-one-document-out average precision of $P(\text{unfair} \mid \text{theme combination})$, scored against the CLAUDETTE unfairness labels, which are never projected. T10 merges across unfairness strata; T14 and T11 do not.

Taxonomy	Classes	α -MASI	Nominal α	Δ vs. T20 [95% CI]	Signal loss
T20	20	0.658	0.732	—	—
T14	14	0.693	0.768	+0.035 [0.030, 0.042]	6.4%
T11	11	0.725	0.794	+0.067 [0.058, 0.077]	5.9%
T10	10	0.737	0.800	+0.078 [0.068, 0.090]	13.2%

Table 2. Agreement with the frozen gold standard, for all three kinds of system, measured by the same procedure. Human agreement is leave-one-annotator-out, so that each annotator is scored against a gold to which they did not contribute. Judge labels were produced once and stored. Reported, not demonstrated: see Section 4.

System	Setting	κ	Accuracy
Human annotators	leave-one-annotator-out	0.859	0.871
Best LLM judge	stored labels, T20	0.581	0.601
Legal-BERT, fine-tuned	T20, 5-fold by document	0.695	—
Legal-BERT, fine-tuned	T11, 5-fold by document	0.720	—

The effect of granularity. Consolidating the taxonomy recovers agreement of the same order as the format costs. Merging T20 into T11 raises α -MASI by 0.067 [0.058, 0.077], carrying the layer across the 0.667 threshold conventionally used for tentative conclusions [25] — a threshold the annotation taxonomy itself does not reach. Two properties make this more than a mechanical artefact of coarser categories. First, the gain *replicates out of sample*: the 17 contracts held out from the design of the merges yield +0.069, against +0.066 on the 33 contracts used to design them. Second, T11 *dominates* the intermediate T14: it gains twice as much agreement (+0.067 against +0.035) while losing *less* downstream signal (5.9% against 6.4%). Granularity is therefore a genuine lever on reliability, and not merely a redistribution of the same disagreements over fewer categories. What the result does not establish is that the merged categories are legally better: it measures how consistently annotators apply them, not whether they carve the domain well.

The legal constraint on consolidation. Consolidation cannot be driven by agreement alone. T10, obtained by merging along measured confusion clusters without regard to the legal status of the clauses, buys a further 0.012 of α -MASI over T11 — and loses 13.2% of the unfairness signal instead of 5.9%, a yield of 0.59 points of agreement per unit of signal against 1.13 for T11. The merge responsible joins two themes whose observed unfairness rates fall in opposite strata, 0.385 against 0.039 — a distinction consumer protection law draws [36,37] and that the agreement signal does not express. A purely statistical consolidation selects exactly this taxonomy; the legal constraint rules it out. Reliability gains and downstream utility are thus not interchangeable, and the boundary between them is drawn by law, not by data.

Reported: the human ceiling, the judges and a fine-tuned encoder. Against the frozen gold, trained annotators reach $\kappa = 0.859$ in leave-one-annotator-out, the best of the four LLM judges reaches 0.581, and a fine-tuned Legal-BERT reaches 0.720 on T11 (Table 2). We report these figures rather than demonstrate them (Section 5). Their ordering is nonetheless unambiguous: a compact, domain-specific encoder outperforms every judge while remaining well below the human ceiling.

5. Discussion and Limitations

Two consequences follow for annotation practice. First, the reliability cost of multi-label annotation need not be paid by abandoning multi-label annotation: it can be recovered by consolidating the taxonomy, which preserves the ability to record that a clause does two things at once. Second, that consolidation cannot be delegated to the agreement signal. On this corpus, a purely data-driven procedure selects precisely the taxonomy that collapses the legal distinction the downstream task depends on, so the criterion has to come from outside the annotation statistics [36,37]. For the evaluation of language models, the gap between the human ceiling and the best judge (0.859 against 0.581) is large enough that judges are not substitutable for trained annotators on this task [23,22]; a compact domain-specific encoder [12,13] closes much, though not all, of that gap.

Limitations. The four judges we evaluate are the four that assisted annotation, so the benchmark is not free of circularity [20]: the engine favoured none of them and a third of gold labels match no proposal, but an error shared by all four could survive. Excluding the strongest judge leaves the second at $\kappa = 0.515$, which widens the gap to the human ceiling rather than narrowing it. The layer covers one legal system and one language. Judge labels were produced once and stored, so the benchmark measures a historical output rather than the current state of the art. The fine-tuned encoder is a single model with early stopping on the test fold, making its reported agreement mildly optimistic. Finally, the aggregated gold carries only 1.04 labels per sentence, so a multi-label score computed on it would not measure the difficulty of the task as annotated; the multi-label cost reported here is measured on individual votes, where the secondary themes survive.

6. Conclusion and Availability

A thematic layer over CLAUDETTE makes it possible to ask what a clause is about, and not only whether it is unfair, and to measure how reliably it can be answered. The cost of the format and the effect of granularity are separable and of different orders, and the consolidation that agreement alone would favour is ruled out by a legal constraint no statistical criterion expresses.

Availability. We release the layer with its individual votes, arbitration trail and frozen gold standard, the codebook and protocol, the frozen taxonomy projections with the fingerprint under which every figure here was computed, and the annotation platform used to produce it [38,39,40]. The layer can be browsed and downloaded from the running instance at <https://pactiva.legal>, and the data and platform source are also at <https://github.com/elazharjebbari/claire-studio>.

References

- [1] Lippi M, Palka P, Contissa G, Lagioia F, Micklitz HW, Sartor G, et al. CLAUDETTE: an automated detector of potentially unfair clauses in online terms of service. *Artificial Intelligence and Law*. 2019;27(2):117-39.
- [2] Lagioia F, Ruggeri F, Drazewski K, Lippi M, Micklitz HW, Torroni P, et al. Deep learning for detecting and explaining unfairness in consumer contracts. In: *Legal Knowledge and Information Systems (JURIX 2019)*. vol. 322 of *Frontiers in Artificial Intelligence and Applications*; 2019. p. 43-52.
- [3] Braun D, Matthes F. Clause topic classification in German and English standard form contracts. In: *Proceedings of the Fifth Workshop on e-Commerce and NLP (ECNLP)*; 2022. p. 199-209.
- [4] Löffler C, Martínez Freile A, Rey Pizarro T. Predicting potentially abusive clauses in Chilean terms of services with natural language processing. *Artificial Intelligence and Law*. 2025.
- [5] Bayerl PS, Paul KI. What determines inter-coder agreement in manual annotations? A meta-analytic investigation. *Computational Linguistics*. 2011;37(4):699-725.
- [6] Marchal M, Scholman M, Yung F, Demberg V. Establishing annotation quality in multi-label annotations. In: *Proceedings of the 29th International Conference on Computational Linguistics (COLING)*; 2022. p. 3659-68.
- [7] Ruggeri F, Lagioia F, Lippi M, Torroni P. Detecting and explaining unfairness in consumer contracts through memory networks. *Artificial Intelligence and Law*. 2022;30(1):59-92.
- [8] Drawzeski K, Galassi A, Jabłonowska A, Lagioia F, Lippi M, Micklitz HW, et al. A corpus for multilingual analysis of online terms of service. In: *Proceedings of the Natural Legal Language Processing Workshop (NLLP)*; 2021. p. 1-8.
- [9] Galassi A, Lagioia F, Jabłonowska A, Lippi M. Unfair clause detection in terms of service across multiple languages. *Artificial Intelligence and Law*. 2024.
- [10] Liepiņa R, Ruggeri F, Lagioia F, Lippi M, Drazewski K, Torroni P. Explaining potentially unfair clauses to the consumer with the CLAUDETTE tool. In: *Proceedings of the Natural Legal Language Processing Workshop (NLLP)*. vol. 2645 of *CEUR Workshop Proceedings*; 2020. .
- [11] Chalkidis I, Jana A, Hartung D, Bommarito M, Androutsopoulos I, Katz DM, et al. LexGLUE: a benchmark dataset for legal language understanding in English. In: *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics*; 2022. p. 4310-30.
- [12] Chalkidis I, Fergadiotis M, Malakasiotis P, Aletas N, Androutsopoulos I. LEGAL-BERT: the muppets straight out of law school. In: *Findings of the Association for Computational Linguistics: EMNLP 2020*; 2020. p. 2898-904.
- [13] Dominguez-Olmedo R, Nanda V, Abebe R, Bechtold S, Engel C, Frankenreiter J, et al. Lawma: the power of specialization for legal annotation. In: *International Conference on Learning Representations (ICLR)*; 2025. .
- [14] Panarelli M, Galassi A, Lagioia F, Liepiņa R, Lippi M, Palka P, et al. Is it worth using LLMs for unfair clause detection in terms of service? In: *Proceedings of the 20th International Conference on Artificial Intelligence and Law (ICAIL)*; 2025. .
- [15] Frasheri M, Bakhtiarnia A, Esterle L, Iosifidis A. Are LLM-based methods good enough for detecting unfair terms of service?; 2024. [ArXiv:2409.00077](https://arxiv.org/abs/2409.00077).
- [16] Savelka J, Ashley KD. The unreasonable effectiveness of large language models in zero-shot semantic annotation of legal texts. *Frontiers in Artificial Intelligence*. 2023;6:1279794.
- [17] Gilardi F, Alizadeh M, Kubli M. ChatGPT outperforms crowd workers for text-annotation tasks. *Proceedings of the National Academy of Sciences*. 2023;120(30):e2305016120.
- [18] Schepers I, Bruijn M, Wieling M, Vols M. The price of automated case law annotation: comparing the cost and performance of GPT-4o and student annotators. *Artificial Intelligence and Law*. 2025.
- [19] Li M, Shi T, Ziemis C, Kan MY, Chen NF, Liu Z, et al. CoAnnotating: uncertainty-guided work allocation between human and large language models for data annotation. In: *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*; 2023. p. 1487-505.
- [20] Choi AS, Akter SS, Singh JP, Anastasopoulos A. The LLM effect: are humans truly using LLMs, or are they being influenced by them instead? In: *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*; 2024. p. 22032-54.
- [21] Zheng L, Chiang WL, Sheng Y, Zhuang S, Wu Z, Zhuang Y, et al. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. In: *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*; 2023. .

- [22] Bavaresco A, Bernardi R, Bertolazzi L, Elliott D, Fernández R, Gatt A, et al. LLMs instead of human judges? A large scale empirical study across 20 NLP evaluation tasks. In: Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Short Papers); 2025. p. 238-55.
- [23] Thalken R, Stiglitz EH, Mimno D, Wilkens M. Modeling legal reasoning: LM annotation at the edge of human agreement. In: Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing; 2023. p. 9252-65.
- [24] Krippendorff K. Content Analysis: An Introduction to Its Methodology. 4th ed. Sage; 2018.
- [25] Passonneau RJ. Measuring agreement on set-valued items (MASI) for semantic and pragmatic annotation. In: Proceedings of the Fifth International Conference on Language Resources and Evaluation (LREC); 2006. .
- [26] Artstein R, Poesio M. Inter-coder agreement for computational linguistics. *Computational Linguistics*. 2008;34(4):555-96.
- [27] Gwet KL. Computing inter-rater reliability and its variance in the presence of high agreement. *British Journal of Mathematical and Statistical Psychology*. 2008;61(1):29-48.
- [28] Tuggener D, von Däniken P, Peetz T, Cieliebak M. LEDGAR: a large-scale multi-label corpus for text classification of legal provisions in contracts. In: Proceedings of the 12th Language Resources and Evaluation Conference (LREC); 2020. p. 1235-41.
- [29] Hendrycks D, Burns C, Chen A, Ball S. CUAD: an expert-annotated NLP dataset for legal contract review. In: Advances in Neural Information Processing Systems, Datasets and Benchmarks Track; 2021. .
- [30] Wilson S, Schaub F, Dara AA, Liu F, Cherivirala S, Giovanni Leon P, et al. The creation and analysis of a website privacy policy corpus. In: Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics; 2016. p. 1330-40.
- [31] Braun D, Matthes F. AGB-DE: a corpus for the automated legal assessment of clauses in German consumer contracts. In: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics; 2024. .
- [32] Braun D. I beg to differ: how disagreement is handled in the annotation of legal machine learning data sets. *Artificial Intelligence and Law*. 2024;32:839-62.
- [33] Plank B. The “problem” of human label variation: on ground truth in data, modeling and evaluation. In: Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing; 2022. p. 10671-82.
- [34] Uma AN, Fornaciari T, Hovy D, Paun S, Plank B, Poesio M. Learning from disagreement: a survey. *Journal of Artificial Intelligence Research*. 2021;72:1385-470.
- [35] Nangia N, Bowman SR. Human vs. Muppet: a conservative estimate of human performance on the GLUE benchmark. In: Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics; 2019. p. 4566-75.
- [36] Council of the European Union. Council Directive 93/13/EEC of 5 April 1993 on unfair terms in consumer contracts; 1993. Official Journal of the European Communities, L 95, 21 April 1993.
- [37] Micklitz HW, Palka P, Panagis Y. The empire strikes back: digital control of unfair terms of online services. *Journal of Consumer Policy*. 2017;40:367-88.
- [38] Stenetorp P, Pyysalo S, Topić G, Ohta T, Ananiadou S, Tsujii J. brat: a web-based tool for NLP-assisted text annotation. In: Proceedings of the Demonstrations at EACL; 2012. p. 102-7.
- [39] Klie JC, Bugert M, Boullosa B, Eckart de Castilho R, Gurevych I. The INCEpTION platform: machine-assisted and knowledge-oriented interactive annotation. In: Proceedings of COLING 2018, System Demonstrations; 2018. p. 5-9.
- [40] Pei J, Ananthasubramaniam A, Wang X, Zhou N, Sargent J, Dedeloudis A, et al. POTATO: the portable text annotation tool. In: Proceedings of EMNLP 2022, System Demonstrations; 2022. .